Accuracy Is Not Enough: Position Bias, Length Bias and Calibration in Pairwise Preference Models for LLM Responses

Ashish Sinha
Indian Institute of Technology Bhilai
ashishsinha@iitbhilai.ac.in

Abstract

Pairwise preference models decide which of two LLM responses a human would prefer; they are the reward models of RLHF and the judges of automatic evaluation. They are usually compared by accuracy alone. We build a cascade of preference models, from a length-only baseline through TF-IDF, BiLSTM, BERT and DeBERTa-v3 cross-encoders and a bi-encoder to a few-shot LLaMA-3-8B judge, and audit them for position bias, length bias, calibration, agreement with human vote distributions and domain shift. On Anthropic HH-RLHF, a 184M-parameter DeBERTa ensemble beats the 8B LLM judge by 4.7 points (66.5% vs. 61.9%) and is far better calibrated (ECE 0.04 vs. 0.31). Every cross-encoder changes its decision on 14–23% of pairs when the two responses are swapped; training-time swap augmentation does not fix this for transformers, while averaging both orders at inference restores consistency and adds 1 point of accuracy. All architectures prefer the longer of two same-content answers 59–68% of the time. On Stanford Human Preferences, training on vote shares instead of majority labels reduces calibration error 3–4.5$\times$ and the divergence from human vote distributions by 36–60% at every model size, without hurting accuracy; post-hoc temperature scaling matches soft labels on top-label calibration but not on the vote distribution. Neither MC dropout nor deep ensembles detect pairs on which humans disagree, and helpfulness-trained models fall below chance on harmlessness prompts.

1 Introduction

Preference models are trained on pairs of responses to the same prompt with a human judgement of which is better (Christiano et al., 2017; Stiennon et al., 2020). They supply the reward signal for RLHF (Ouyang et al., 2022; Bai et al., 2022), are optimized implicitly by direct preference methods (Rafailov et al., 2023), and are increasingly replaced or complemented by prompted LLM judges (Zheng et al., 2023). Benchmarks rank them by pairwise accuracy (Lambert et al., 2024). Accuracy, however, says nothing about whether a model's decision depends on the order in which the responses are shown, whether it rewards length rather than content, whether its probabilities can be trusted, or whether it knows when humans themselves disagree. Each of these failure modes propagates into whatever system the preference model supervises.

We therefore treat a preference model as an instrument to be audited. We ask three research questions: **RQ1** Can a small fine-tuned encoder predict human preferences as well as a much larger prompted LLM judge? **RQ2** How sensitive are preference models to non-content factors (presentation order and response length), and which mitigations work? **RQ3** Does training on annotator vote distributions (soft labels) instead of majority labels improve calibration and agreement with human disagreement? We additionally test whether model uncertainty flags human disagreement and how models transfer across domains and label sources.

Our contributions are: (1) a single evaluation protocol applied to seven model tiers, with paired-bootstrap significance tests for every comparison; (2) evidence that position bias in cross-encoders is not removed by swap augmentation but is removed, with an accuracy gain, by symmetrizing predictions at inference, and that a bi-encoder is order-invariant at no accuracy cost; (3) a controlled length-perturbation test showing a persistent preference for longer same-content answers in all architectures; (4) evidence that soft labels improve calibration and distributional agreement at every model size, together with the finding that temperature scaling recovers top-label calibration but not distributional agreement; and (5) negative results: model uncertainty does not track human

disagreement, and helpfulness and harmlessness preferences are anti-correlated across domains.

2 Related Work

Preference and reward models. Reward models are usually trained with a Bradley–Terry objective (Bradley and Terry, 1952) on human comparisons (Christiano et al., 2017; Stiennon et al., 2020; Ouyang et al., 2022). HH-RLHF (Bai et al., 2022) is the standard public source of such comparisons; RewardBench (Lambert et al., 2024) evaluates reward models by accuracy across categories. We complement accuracy with bias, calibration and disagreement measurements.

Biases of automatic judges. LLM judges favour a particular response position and longer responses (Zheng et al., 2023; Wang et al., 2023), and LLMs are sensitive to option order more generally (Pezeshkpour and Hruschka, 2023). Length correlations pervade RLHF reward models (Singhal et al., 2023), motivating length-controlled evaluation (Dubois et al., 2024). We measure the same biases for small fine-tuned models and compare mitigations.

Calibration, uncertainty and disagreement. Modern neural networks are overconfident and can be recalibrated post hoc with temperature scaling (Guo et al., 2017) or isotonic regression (Zadrozny and Elkan, 2002). MC dropout (Gal and Ghahramani, 2016) and deep ensembles (Lakshminarayanan et al., 2017) estimate predictive uncertainty. Training on soft labels that reflect annotator disagreement can improve robustness and calibration (Peterson et al., 2019; Uma et al., 2021). We test these ideas on preference data with real vote distributions (Ethayarajh et al., 2022), and compare human with AI-generated preference labels (Cui et al., 2024).

3 Data

HH-RLHF (Bai et al., 2022). We split each chosen/rejected dialogue into the shared prompt (up to the last assistant turn) and the two final responses, dropping 119 pairs without a shared prefix. The three *helpful* subsets give 111,311 training, 5,997 validation and 6,194 test pairs; validation is split by prompt, and 33 training pairs whose prompt occurs in the test set are removed. The 44,729 *harmless-base* pairs serve as an out-of-domain test set; for the reverse direction we split them by prompt into 38,741 / 1,995 / 3,993 pairs. Each pair has one human label.

Stanford Human Preferences (SHP) (Ethayarajh et al., 2022) contains 385K pairs of Reddit comments to the same post in 18 subreddits. We use the upvote share $p_A = s_A/(s_A + s_B)$ as a soft label. Training pairs pass a quality gate ($|s_A - s_B| > 10$, $s_A + s_B > 20$, $2|p_A - 0.5| \geq 0.2$; 37.9% pass; at most five pairs per post), giving 72,216 pairs. The 18,406-pair test set is *not* gated, so that 3,744 high-disagreement pairs ($0.4 < p_A < 0.6$, 20.3%) remain for evaluation.

UltraFeedback (Cui et al., 2024). For an AI-label ablation we use the binarized GPT-4-judged comparisons, drop ties, and keep 51,344 / 2,000 / 1,715 pairs, with soft target $\sigma(\text{score}_A - \text{score}_B)$.

Preprocessing. In every dataset the preferred response is placed in position A or B uniformly at random (seed 42), so labels carry no position information. All transformer inputs share one token budget: `[CLS]` prompt `[SEP]` A `[SEP]` B `[SEP]`, keeping the last 128 prompt tokens and the first 190 tokens of each response. With the DeBERTa tokenizer, 7.1% of HH and 18.6% of SHP training responses exceed the response budget.

Human length preference. Humans prefer the longer response in 55.8% of HH-helpful pairs and 68.5% of gated SHP pairs (62.8% on the ungated test set), but only 42.0% of HH-harmless pairs (Figure 5 in the appendix). A length-only model is therefore a strong baseline on helpfulness data and a misleading one on harmlessness data.

4 Models

All models output p_A, the probability that response A is preferred.

T0: Length only and TF-IDF. Logistic regression on the log length ratio $\log \frac{|A|+1}{|B|+1}$, and on the difference of TF-IDF vectors of the two responses (10K uni- and bigrams). Neither has an intercept, so both are order-invariant by construction.

T1: BiLSTM. A two-layer bidirectional LSTM (Hochreiter and Schmidhuber, 1997) with attention pooling (10.7M parameters, trained from scratch on the BERT WordPiece vocabulary), either over the concatenated input (*cross*) or applied separately to (prompt, A) and (prompt, B) with logit $= s(A) - s(B)$ (*siamese*).

T2/T3: Cross-encoders. BERT-base (Devlin et al., 2019) (110M) and DeBERTa-v3-base (He et al., 2023) (184M) read the full input; a linear layer on the `[CLS]` state gives the logit. Both responses attend to each other, which also allows the model to learn a position preference.

T5: Bi-encoder. A shared BERT-base scores `[CLS]` prompt `[SEP]` response `[SEP]` for each response separately and $p_A = \sigma(s(A) - s(B))$, the standard reward-model design (Ouyang et al., 2022), as in Siamese sentence encoders (Reimers and Gurevych, 2019). Swapping A and B exactly negates the logit.

T6: LLM judge. LLaMA-3-8B-Instruct (Dubey et al., 2024), not fine-tuned, is shown the conversation and both responses with three HH-training examples whose answers alternate A, B, A, and asked which response is better (prompt in Appendix B). p_A is the softmax over the next-token logits of "A" and "B".

Training. Neural models minimize binary cross-entropy against the hard label or the soft target p_A. Transformers use AdamW (learning rate 2×10^{-5}, weight decay 0.01, 10% warmup, linear decay), an effective batch of 32 and at most three epochs, with validation every 1,000 steps and early stopping on validation loss (patience 3), which selects the best-calibrated checkpoint. BERT trains in bf16; DeBERTa-v3 trains in fp32 because bf16 autocast produced NaN predictions. Hyperparameters are listed in Appendix A. Main runs use seed 42; deep ensembles add seeds 1 and 2. Total compute was about 35 GPU-hours on one RTX 6000 Ada.

5 Evaluation Protocol

Accuracy and calibration. We report accuracy (with 95% bootstrap intervals), AUC, NLL, and expected calibration error (ECE) over the confidence of the predicted winner (15 bins on $[0.5, 1]$). On SHP we also report the Bernoulli KL divergence from the human vote share to the prediction, $\mathrm{KL}(p_A^{\text{human}} \,\|\, p_A^{\text{model}})$, overall and on the high-disagreement subset.

Swap test (position bias). Every pair is scored in both orders. *Swap consistency* is the fraction of pairs whose decision follows the swap; *position bias* is the mean of $p_{AB} + p_{BA} - 1$ (positive values favour the first slot). *Symmetrization* averages the two orders, $p_{\text{SYM}} = (p_{AB} + 1 - p_{BA})/2$.

Test B (length perturbation). From 500 HH test pairs whose chosen response has 40–150 words, Qwen2.5-7B-Instruct (Qwen Team, 2024) rewrites the chosen response to be about 40% shorter while keeping its content (achieved median length ratio 0.46). We evaluate five variants with randomized order: the original pair; the compressed chosen response against the rejected one; the rejected response padded with two content-free sentences; and two unlabelled same-content pairs (original vs. compressed, original vs. padded), on which we measure how often the model picks the longer version.

Calibration and uncertainty. Temperature scaling, isotonic regression and symmetrization are fitted on each model's own validation set. MC dropout uses 10 stochastic passes; deep ensembles average the logits of three seeds. We measure with AUROC whether the spread across passes or members, or low confidence $|p_A - 0.5|$, identifies errors and high-disagreement SHP pairs.

Significance. Comparisons on the same test pairs use a paired bootstrap with 2,000 resamples (Efron and Tibshirani, 1994); we report the difference, its 95% interval and a two-sided p-value (Appendix C).

6 Results

6.1 RQ1: Small fine-tuned models vs. an LLM judge

Accuracy rises steadily with model capacity (Table 1, Figure 1): from 55.6% for response length alone to 60.3% for TF-IDF, 63.0% for the best BiLSTM, 63.6% for BERT and 65.8% for DeBERTa-v3 (DeBERTa vs. BERT $+2.2$ points, $p = 0.003$). Seeds vary by about ± 0.5 points (BERT 63.6/64.4/63.6, DeBERTa 65.8/66.2/66.4), and a three-seed DeBERTa ensemble reaches 66.5%, or 67.5% when symmetrized.

The few-shot LLaMA-3-8B judge reaches 61.9%, between TF-IDF and the BiLSTM. The DeBERTa ensemble is 4.7 points more accurate (95% CI [3.2, 6.1], $p < 0.001$), and the gap remains 3.9 points when both models are symmetrized. The judge is also badly calibrated: ECE 0.313 against 0.039, and temperature scaling needs $T = 11.1$ to reduce its ECE to 0.013. Its

Tier	Model	Params	Accuracy [95% CI]	Acc. SYM	AUC	ECE	Swap cons.	Harmless
T0	Length only	1	55.6 [54.3, 56.8]	55.6	.590	.020	.98	42.2
T0	TF-IDF + LR	10K	60.3 [59.1, 61.6]	60.3	.656	.016	1.00	38.5
T1	BiLSTM cross	11M	61.7 [60.6, 62.9]	62.2	.676	.024	.78	37.3
T1	BiLSTM siamese	11M	63.0 [61.9, 64.3]	63.0	.686	.012	1.00	36.1
T2	BERT cross	110M	63.6 [62.4, 64.8]	64.6	.706	.022	.77	35.2
T2	BERT cross, 3-seed ensemble	330M	64.4 [63.2, 65.8]	65.1	.714	.024	.82	34.6
T5	BERT bi-encoder	110M	64.6 [63.4, 65.9]	64.6	.711	.023	1.00	34.3
T3	DeBERTa-v3 cross	184M	65.8 [64.6, 67.0]	66.8	.726	.064	.82	34.6
T3	DeBERTa-v3 cross, 3-seed ensemble	552M	**66.5** [65.4, 67.7]	**67.5**	**.736**	.039	.86	33.8
T6	LLaMA-3-8B judge, 3-shot	8B	61.9 [60.8, 63.1]	63.6	.670	.313	.60	**52.6**†

Table 1: HH-RLHF helpful test set (6,194 pairs), all models trained on HH helpful. Acc. SYM: accuracy after averaging both presentation orders. Swap cons.: fraction of decisions that follow an A/B swap. Harmless: accuracy on the 44,729 harmless pairs (†1,000-pair random sample for the judge).

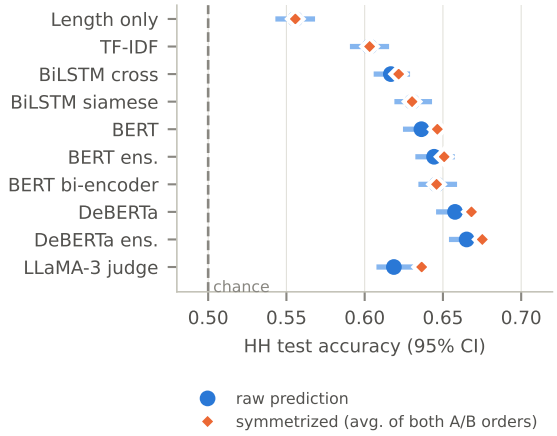


Figure 1: HH test accuracy by model with 95% bootstrap intervals, before (circles) and after (diamonds) symmetrizing both presentation orders.

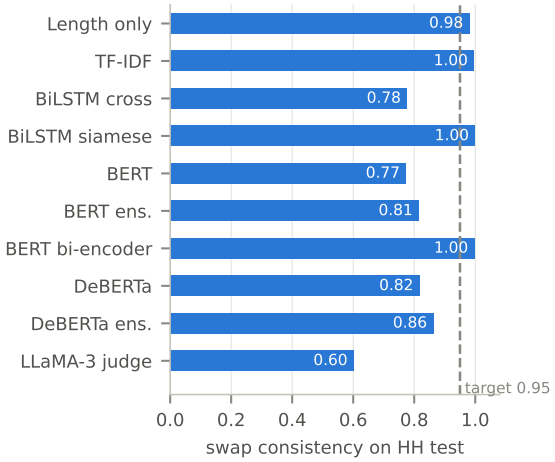


Figure 2: Swap consistency on HH test (1.0 = the decision never depends on presentation order); dashed line: the project's 0.95 target.

one advantage is robustness to domain shift (§6.5). **Answer to RQ1:** on in-distribution human preferences a fine-tuned 184M-parameter encoder is significantly more accurate and far better calibrated than an 8B prompted judge. Our initial target of 72% accuracy was not reached by any model; HH labels come from a single annotator, which limits attainable accuracy.

6.2 RQ2: Position and length bias

Position. Order-invariant models (length, TF-IDF, siamese BiLSTM, bi-encoder) are consistent under swaps by construction. Every model that reads both responses jointly is not (Figure 2): the BiLSTM cross model and BERT follow the swap on only 77–78% of pairs, DeBERTa on 82%, and the LLM judge on 60%, with a strong preference for the first response (position bias $+0.31$). Under domain shift the problem grows: SHP-trained BERT and DeBERTa follow the swap on only 43–46% of HH pairs.

Training on randomly swapped copies (swap probability 0.5) raises the BiLSTM's consistency from 0.78 to 0.88 but barely changes the transformers (BERT 0.77→0.79, accuracy $+0.8$ points, $p = 0.10$; DeBERTa 0.82→0.80). Symmetrizing at inference, in contrast, yields consistency 0.999 and *improves* accuracy (BERT $+1.0$, $p = 0.024$; DeBERTa $+1.1$, $p = 0.009$), at the cost of a second forward pass. The bi-encoder is consistent by design and at least as accurate as the BERT cross-encoder ($+1.0$ points, $p = 0.054$): cross-attention between the responses buys no accuracy on HH, only position sensitivity.

Length. On HH test the correlation between length difference and the label is $r = 0.11$ for humans, but 0.23–0.41 for the trained models (TF-IDF 0.41, BERT 0.35, DeBERTa 0.23) and 0.78 for the length-only model. Test B isolates the effect (Table 2, Figure 3). When the better answer

Model	Orig.	Compr.	Longer	Padded
BERT cross	75.2	68.0	66	41
BERT + swap aug.	71.2	68.0	59	50
BERT no prompt	73.0	69.6	61	44
BERT bi-encoder	74.2	67.2	68	44
DeBERTa cross	74.4	68.8	63	55
DeBERTa + swap aug.	73.0	65.6	63	48

Table 2: Test B on 500 HH pairs. Orig./Compr.: accuracy on the original pairs and with the preferred answer compressed to about half its length. Longer/Padded: % of same-content pairs in which the model prefers the longer (uncompressed) or the filler-padded version; 50 = no preference.

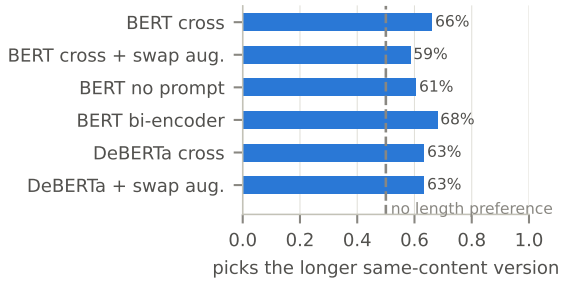


Figure 3: Share of same-content pairs in which each model picks the longer version (Test B).

is compressed to about half its length, accuracy drops by 3–7 points; given two versions of the same answer, every model prefers the longer one 59–68% of the time. Content-free padding, by contrast, is not rewarded (41–55%), so the bias is towards more *content-bearing* text rather than towards more tokens. Accuracy on compressed pairs stays at 66–70%, above the project's 65% target. With 500 pairs, differences between models in Table 2 are within sampling error (± 4 points). **Answer to RQ2:** position bias is large in joint-encoding models and is best removed at inference or by architecture; length bias persists in every architecture.

6.3 RQ3: Soft labels and calibration

Table 3 and Figure 4 compare training on majority labels with training on vote shares. At every tier, soft labels reduce ECE 3–4.5× (BiLSTM 0.068→0.021, BERT 0.062→0.021, DeBERTa 0.109→0.024) and the KL divergence from human votes by 39%, 36% and 60%, with even larger reductions on high-disagreement pairs (all $p < 0.001$). Accuracy is unchanged or higher (BERT +0.7 points, $p = 0.008$; BiLSTM and DeBERTa n.s.). The hard-label DeBERTa is the most overconfident model in the study (calibration temperature $T = 1.91$); larger models overfit hard labels more.

	Labels	Acc.	ECE	KL	ECE_{HD}	KL_{HD}
BiLSTM	hard	66.3	.068	.206	.102	.183
	hard + T	66.3	.014	.133	.033	–
	soft	66.6	.021	**.126**	.033	**.063**
BERT	hard	68.0	.062	.205	.098	.189
	hard + T	68.0	.014	.137	.045	.092
	soft	**68.7**	.021	**.131**	.046	**.086**
DeBERTa	hard	69.9	.109	.339	.153	.384
	hard + T	69.9	.015	.148	.051	.122
	soft	70.1	.024	**.137**	.048	**.105**

Table 3: SHP test set (18,406 pairs). Hard: trained on majority labels; hard + T: the same model after temperature scaling; soft: trained on vote shares. HD: the 3,744 high-disagreement pairs ($0.4 < p_A < 0.6$). KL: divergence from the human vote share.

The relevant baseline, however, is not the raw hard-label model but the same model after temperature scaling. Temperature scaling achieves slightly *lower* top-label ECE than soft labels (BERT 0.014 vs. 0.021, $p = 0.033$; DeBERTa 0.015 vs. 0.024, $p = 0.008$), but soft labels remain significantly closer to the human vote distribution (KL 0.131 vs. 0.137 for BERT and 0.137 vs. 0.148 for DeBERTa; on high-disagreement pairs 0.086 vs. 0.092 and 0.105 vs. 0.122; all $p < 0.001$). Soft-label models are slightly underconfident at the highest confidence levels (Figure 4). **Answer to RQ3:** soft labels improve calibration and distributional agreement at every model size; if only top-label calibration matters, temperature scaling is sufficient, but matching how humans split requires training on the split.

6.4 Does model uncertainty flag human disagreement?

A practical use of uncertainty is to route ambiguous pairs to human review. We test whether MC dropout spread, ensemble spread, or low confidence identifies the 20.3% of SHP test pairs on which voters split (Table 4). None does: AUROC is 0.50–0.51 for hard-label models and 0.52–0.54 for soft-label models, and flagging predictions in the 0.4–0.6 band raises the share of high-disagreement pairs only from 20.3% to 21–22%. Uncertainty is somewhat more informative about the model's own errors in domain (AUROC 0.58–0.59 for DeBERTa on HH; 0.62 for soft-label models on SHP). Model uncertainty and human disagreement are different quantities, and the former

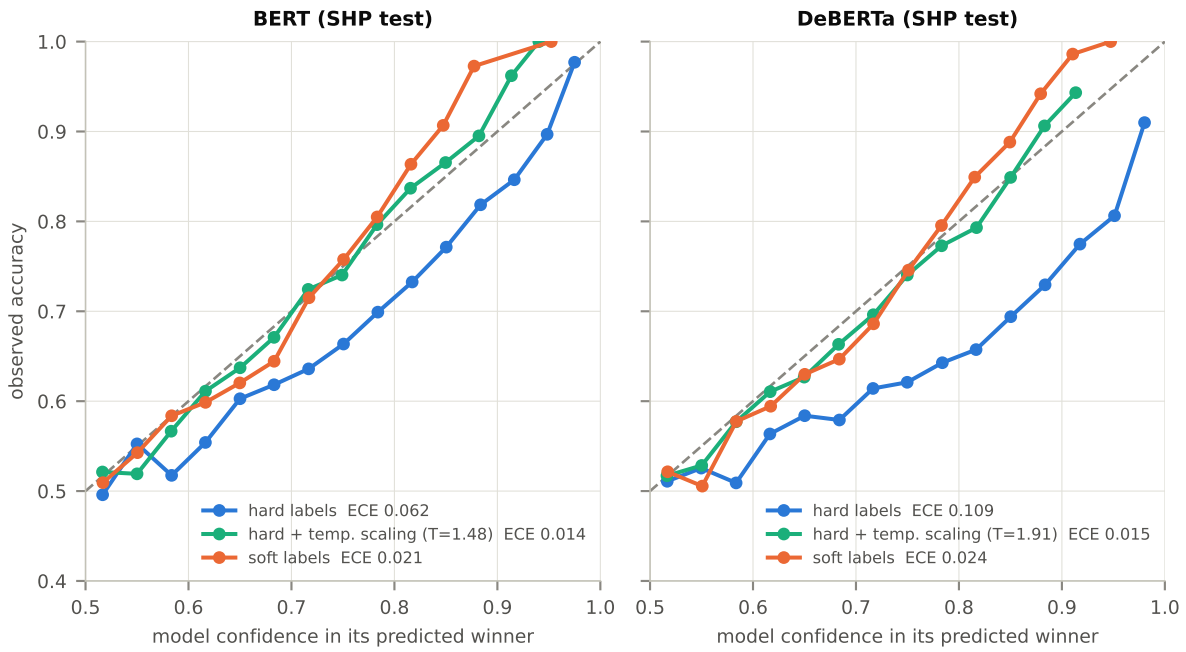


Figure 4: Reliability diagrams on SHP test for BERT and DeBERTa trained on hard labels, the same models after temperature scaling, and models trained on soft labels. The dashed diagonal is perfect calibration.

Model, method	Errors	HD (spread)	HD ($\lvert p_A - .5\rvert$)
BERT hard, MC dropout	.506	.503	.508
BERT hard, ensemble	.506	.513	.507
DeBERTa hard, MC dropout	.502	.499	.509
DeBERTa hard, ensemble	.520	.496	.499
BERT soft, MC dropout	.621	.523	.539
DeBERTa soft, MC dropout	.618	.521	.534

Table 4: AUROC on SHP test for detecting the model's errors and high-disagreement (HD) pairs from the spread across dropout passes or ensemble members, or from low confidence. 0.5 = no signal. Hard models were trained on HH, soft models on SHP.

Trained on ↓ / tested on →	Helpful	Harmless	SHP	UF
HH helpful (human)	**63.6**	35.0	55.0	50.4
HH harmless (human)	42.0	**69.6**	44.6	49.0
SHP (human, soft)	54.0	–	**68.7**	–
UltraFeedback (GPT-4)	55.2	44.7	57.8	**65.7**

Table 5: Transfer matrix for the BERT cross-encoder (accuracy). Harmless: the held-out harmless test split (3,993 pairs). UF: UltraFeedback test set.

cannot stand in for the latter.

6.5 Domain shift and label source

Helpfulness-trained models score 34–42% on harmlessness prompts, *below* chance, and the stronger the model, the lower the score (length-only 42.2%, DeBERTa ensemble 33.8%; Figure 6 in the appendix). The reverse also holds: a model trained on harmless pairs reaches 69.6% on harmless but 42.0% on helpful prompts (Table 5). The two labelling criteria are not just different but opposed: on harmlessness prompts humans tend to prefer the shorter, refusing response, while helpfulness models learned to reward longer, more forthcoming answers. The LLM judge, which was never trained on either, is the only model above chance on harmless prompts (52.6%).

Training on GPT-4 judgements (UltraFeedback) instead of human labels transfers slightly better to unseen human data (SHP 57.8% vs. 55.0% for the HH-trained model, $p < 0.001$) but is much worse on HH human preferences (55.2% vs. 63.6%, $p < 0.001$) and worse calibrated (ECE on HH 0.124 vs. 0.022). AI-generated preferences are not a drop-in substitute for in-domain human labels.

6.6 Ablation: is the prompt used?

A BERT cross-encoder that never sees the prompt is as accurate as the full model (63.7% vs. 63.6%, $p = 0.88$). On HH, the model mostly judges response quality in isolation rather than whether a response fits its prompt. Because HH prompts are often generic, this result may not hold for tasks where correctness depends on the question.

7 Discussion

Three practical recommendations follow. *First*, report swap consistency next to accuracy: a modest average position bias (BERT: $+0.06$) hides 23% of decisions that flip with order, so part of a cross-encoder's accuracy rests on order-dependent decisions. Symmetrizing at inference or using a bi-encoder removes the problem without losing accuracy; augmentation alone does not. *Second*, when vote distributions exist, train on them: soft labels give better-calibrated and more human-like probabilities at no cost. When they do not, temperature scaling repairs top-label calibration, including for LLM judges, whose raw probabilities are almost useless (ECE 0.31). *Third*, do not use model uncertainty as a proxy for human disagreement, and do not assume a preference model transfers between helpfulness and harmlessness: the two are anti-correlated, which argues for separate or conditioned reward models.

Length bias is the one failure we could not mitigate. Every architecture prefers longer same-content answers even though the human length correlation on HH is small ($r = 0.11$). Because the bias survives order-invariant architectures, augmentation and the removal of the prompt, it likely reflects the training labels' weak length preference amplified by the model, and calls for length-controlled objectives or counterfactual training data.

8 Conclusion

We audited preference models from a length baseline to an 8B LLM judge on accuracy, position bias, length bias, calibration, disagreement and domain shift. Small fine-tuned encoders are more accurate and much better calibrated than a prompted LLM judge on in-distribution data, but they inherit position and length biases that accuracy hides. Inference-time symmetrization, order-invariant architectures and soft-label training each fix a specific reliability problem; length bias and the gap between model uncertainty and human disagreement remain open.

Limitations

Most configurations were trained with a single seed; the three-seed runs suggest seed variation of about ± 0.5 points, comparable to some reported differences. We used base-size encoders only: DeBERTa-v3-large was not trained because fp32 training would have required about a GPU-day. The 190-token response budget truncates 18.6% of SHP responses, so some length differences are invisible to the encoders. SHP upvote shares are a proxy for annotator votes and are affected by comment timing and visibility. The Test B compressions were produced by an LLM and only spot-checked, and the test has 500 pairs. The LLM judge results cover one model, one prompt and three examples; its harmless and SHP numbers use 1,000-pair samples, and a frontier judge such as GPT-4 was not evaluated. HH labels come from a single annotator per pair, which caps attainable accuracy and prevents measuring disagreement there. No new human annotations were collected, and all data are English.

Ethics Statement

HH-RLHF contains offensive and harmful content, including in its harmlessness subset; we used it only for training and evaluation, and the accompanying demo warns users before showing examples. All datasets are public and contain no personal information beyond public Reddit comments in SHP. Preference models trained on these data encode the values of their annotators and should not be deployed to judge content without review; our results on domain shift show how differently they behave outside their training distribution. Experiments used about 35 GPU-hours.

References

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. 2022. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*.

Ralph Allan Bradley and Milton E. Terry. 1952. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345.

Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30.

Ganqu Cui, Lifan Yuan, Ning Ding, Guanming Yao, Bingxiang He, Wei Zhu, Yuan Ni, Guotong Xie, Ruobing Xie, Yankai Lin, Zhiyuan Liu, and Maosong Sun. 2024. UltraFeedback: Boosting language models with scaled AI feedback. In *Proceed-*

ings of the 41st International Conference on Machine Learning.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4171–4186.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2024. Length-controlled AlpacaEval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*.

Bradley Efron and Robert J. Tibshirani. 1994. *An Introduction to the Bootstrap*. Chapman and Hall/CRC.

Kawin Ethayarajh, Yejin Choi, and Swabha Swayamdipta. 2022. Understanding dataset difficulty with $\mathcal{V}$-usable information. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 5988–6008.

Yarin Gal and Zoubin Ghahramani. 2016. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning*, pages 1050–1059.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *The Eleventh International Conference on Learning Representations*.

Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural Computation*, 9(8):1735–1780.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. 2017. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30.

Nathan Lambert, Valentina Pyatkin, Jacob Morrison, L. J. Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. 2024. RewardBench: Evaluating reward models for language modeling. *arXiv preprint arXiv:2403.13787*.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744.

Joshua C. Peterson, Ruairidh M. Battleday, Thomas L. Griffiths, and Olga Russakovsky. 2019. Human uncertainty makes classification more robust. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9617–9626.

Pouya Pezeshkpour and Estevam Hruschka. 2023. Large language models sensitivity to the order of options in multiple-choice questions. *arXiv preprint arXiv:2308.11483*.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992.

Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. 2023. A long way to go: Investigating length correlations in RLHF. *arXiv preprint arXiv:2310.03716*.

Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F. Christiano. 2020. Learning to summarize with human feedback. In *Advances in Neural Information Processing Systems*, volume 33, pages 3008–3021.

Alexandra N. Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. 2023. Large language models are not fair evaluators. *arXiv preprint arXiv:2305.17926*.

Bianca Zadrozny and Charles Elkan. 2002. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 36.

A Hyperparameters

	BiLSTM	BERT	DeBERTa-v3
Learning rate	10^{-3}	2×10^{-5}	2×10^{-5}
Batch (effective)	64	32	16×2
Max. epochs	6	3	3
Warmup	5%	10%	10%
Precision	fp32	bf16	fp32
Early stopping	2 epochs	3 evals	3 evals
Dropout	0.3	0.1	0.1

Table 6: Training hyperparameters. All use AdamW with weight decay 0.01 and gradient clipping at 1.0; transformers are validated every 1,000 optimizer steps. The token budget is 128 prompt and 2×190 response tokens. Bi-encoder: as BERT.

B Prompts

LLM judge. System: *"You are an expert evaluator of AI assistant responses. You will see a conversation between a human and an AI assistant, followed by two candidate final assistant responses, A and B. Decide which response is better."* User: the conversation (last 300 words), Response A and Response B (first 250 words each), followed by *"Which response is better? Answer with a single letter: A or B."* Three HH training examples precede the query as user/assistant turns.

Test B compression. *"Rewrite the following response so that it is about 40% shorter. Keep every piece of information, every instruction and the same tone. Do not add anything new. Output only the rewritten response."* Padding appends two of six content-free sentences such as *"I hope this helps!"* and *"Let me know if you have any other questions."*

C Significance Tests

D Additional Figures

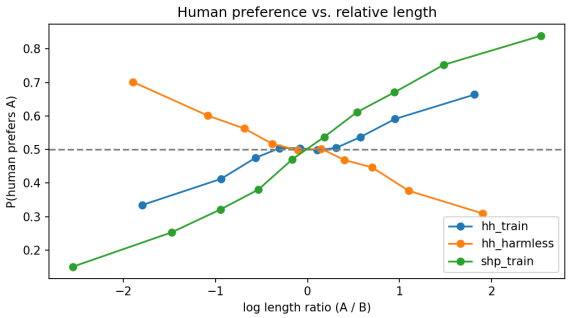


Figure 5: Probability that humans prefer response A as a function of the log length ratio A/B (training sets). Humans prefer longer responses on HH helpful and SHP, but shorter ones on HH harmless.

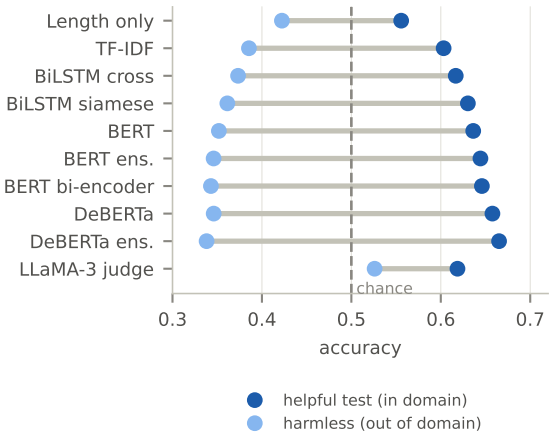


Figure 6: Accuracy of HH-helpful-trained models on the helpful test set and on harmless prompts.

Comparison (A vs. B)	Metric	A	B	Δ [95% CI]	p
DeBERTa ensemble vs. LLaMA-3 judge (HH)	accuracy	.665	.619	+.047 [+.032, +.061]	<.001
DeBERTa ensemble vs. LLaMA-3 judge (HH)	ECE	.039	.313	−.273 [−.284, −.256]	<.001
same, both symmetrized	accuracy	.675	.636	+.039 [+.025, +.053]	<.001
DeBERTa vs. BERT (HH)	accuracy	.658	.636	+.022 [+.009, +.034]	.003
BERT symmetrized vs. raw	accuracy	.646	.636	+.010 [+.002, +.019]	.024
DeBERTa symmetrized vs. raw	accuracy	.668	.658	+.011 [+.003, +.018]	.009
BERT swap augmentation vs. none	accuracy	.645	.636	+.008 [−.001, +.018]	.099
BERT bi-encoder vs. cross-encoder	accuracy	.646	.636	+.010 [+.000, +.020]	.054
BERT without vs. with prompt	accuracy	.637	.636	+.001 [−.008, +.009]	.877
BERT soft vs. hard (SHP)	KL	.131	.205	−.073 [−.076, −.071]	<.001
DeBERTa soft vs. hard (SHP)	KL	.137	.339	−.202 [−.206, −.197]	<.001
BERT soft vs. hard + T (SHP)	ECE	.021	.014	+.007 [+.001, +.014]	.033
BERT soft vs. hard + T (SHP)	KL	.131	.137	−.006 [−.008, −.005]	<.001
DeBERTa soft vs. hard + T (SHP)	ECE	.024	.015	+.009 [+.002, +.014]	.008
DeBERTa soft vs. hard + T (SHP)	KL	.137	.148	−.011 [−.013, −.010]	<.001
Harmless- vs. helpful-trained, on harmless	accuracy	.696	.350	+.346 [+.321, +.370]	<.001
Harmless- vs. helpful-trained, on helpful	accuracy	.420	.636	−.217 [−.236, −.196]	<.001
GPT-4- vs. human-label training, on SHP	accuracy	.578	.550	+.028 [+.018, +.038]	<.001
GPT-4- vs. human-label training, on HH	accuracy	.553	.636	−.084 [−.099, −.068]	<.001

Table 7: Paired bootstrap tests (2,000 resamples) on identical test pairs. The full list, including all SHP metrics, is in `results/significance.csv`.

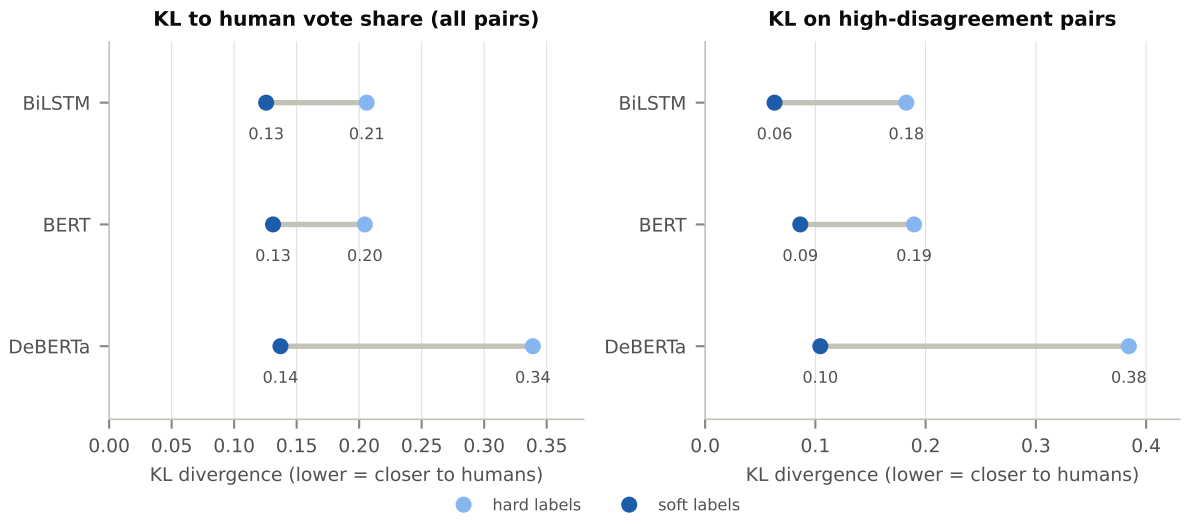


Figure 7: KL divergence from the human vote share on SHP test for hard- and soft-label models, on all pairs and on high-disagreement pairs.